

1. MASTER INVENTORY OF GENERATED PROJECT ARTIFACTS

ARTIFACT FILE PATH	ARTIFACT TITLE	FORMAT / LAYER	CORE ACADEMIC & DEFENSE PURPOSE
docs/ACCOUNTS_CREDENTIALS.md	Master Auth & Credentials	MARKDOWN / AUTH	Directory of all 7 Firebase Auth accounts, passwords (password123), and RBAC gatekeeping passcodes.
docs/PROJECT_DEFENSE_GUIDELINES.md	Oral Defense Playbook	PRESENTATION	5-minute timed presentation script, live demo sequence, examiner trick questions & answers, NDPA 2023 compliance.
docs/DISSERTATION_CORRECTIONS...	Thesis Corrections Guide	DISSERTATION	Page-by-page replacement text transforming the PDF draft from "design-only" to a fully implemented cloud system.
docs/DATABASE_DICTIONARY.md	3NF Data Dictionary	DATABASE DDL	Formal relational data dictionary for MySQL 8.0 & PostgreSQL 16 formatted for direct inclusion into Chapter 4.
docs/PROBLEMS_FEATURES...	Problems & Features Spec	ACADEMIC SPEC	Full academic problem statement, the sweaty finger literature gap, NUC 75% formula, and NDPA 2023 compliance.
diagrams/DIAGRAMS_GALLERY.md	UML & Flowchart Gallery	MERMAID.LIVE	6 standardized Mermaid diagrams (Architecture, Flagged Workflow, NUC 75%, RBAC, ERD, and Multi-Device Sync).
database/schema.sql & seeds.sql	Relational SQL DDL & Seeds	MYSQL / POSTGRES	3NF normalized relational schema scripts including the new CLASS_REP role and 10 pre-loaded student records.
firestore.rules & firebase.json	Cloud Security Rules	FIREBASE CONFIG	Production-grade access control protecting immutable audit logs and restricting flag overrides to faculty and admins.

2. MASTER ACCOUNTS & INSTITUTIONAL GATEKEEPING CREDENTIALS

SYSTEM ROLE	USER NAME	INSTITUTIONAL EMAIL	PASSWORD	PASSCODE	DEFENSE SCENARIO
 Admin	Dr. Kola Balogun	admin@smartbio.edu.ng	password123	GWU-ADMIN-2026	NDPA audit logs, 1-Click Cloud Seeder, SQL export.
 Lecturer	Dr. Olawale Adeyemi	o.adeyemi@smartbio.edu.ng	password123	GWU-FACULTY-2026	Start CSC 401 lecture, stream live radar, resolve flags.
 Lecturer	Prof. Ngozi Okoro	n.okoro@smartbio.edu.ng	password123	GWU-FACULTY-2026	CSC 407 lecture sessions and multi-course moderation.
 Class Rep	Chukwudi Eze	c.eze@student.gwu.edu	password123	GWU-PROCTOR-2026	Proctor tablet scanner kiosk, track live headcount.
 Student (80%)	Benedict Uche (Cleared)	b.uche@student.gwu.edu	password123	(None - Open)	View 80% cleared gauge and mint QR clearance docket.
 Student (100%)	Folake Adebayo	f.adebayo@student.gwu.edu	password123	(None - Open)	100% attendance rate, perfect clearance status.
 Student (50%)	Amina Mohammed (Defaulter)	a.mohammed@student.gwu.edu	password123	(None - Open)	50% attendance, displays automated deficit forecast.

Defense Demonstration Playbook & Chapter 4 Screenshot Matrix

3. 5-STEP LIVE DEMONSTRATION WALKTHROUGH FOR DEFENSE PANEL

Step 1: Role Gatekeeping & Route Guards

- Click **Switch User** → Click **Register / Sign Up**.
- Select **Lecturer** → Passcode field dynamically appears.
- Sign in as **Dr. Olawale Adeyemi** (o.adeyemi@... / password123).
- Highlight:** Navigation bar locks down to Lecturer and Scanner only.

Step 2: Start Cloud-Synchronized Lecture

- Select **CSC 401: Advanced Software Architecture**.
- Click **Start Live Session** → See live timer start.
- Highlight:** Broadcasts immediately to Firebase Firestore `/lecture_sessions` in < 200ms.

Step 3: Simulate Literature Gap (Sweaty Scan)

- Switch to **Scanner Terminal** → Select **Tunde Bakare**.
- Mode: **Sweaty / Blurred Ridge** → Click **Simulate Optical Scan**.
- Highlight:** Warning double-beep plays; HUD logs `FLAGGED EXCEPTION: SWEATY_RIDGE_BLURRED`.

Step 4: Lecturer ID Override (NDPA 2023)

- Switch to **Lecturer Portal** → See Tunde in Flagged Queue.
- Click **Review & Override** → Enter note: *"Verified physical ID"* → Click **Approve & Credit**.
- Highlight:** Attendance credited; immutable audit log created.

Step 5: Student NUC 75% Compliance & Anti-Tamper QR Docket

- Switch to **Student Portal** (b.uche@student.gwu.edu). Show **80% Cleared Progress Gauge**.
- Click **Generate Exam Clearance Docket** → Show official docket with student photo, clearance stamps, and anti-tamper QR code.

4. CHAPTER 4 DISSERTATION SCREENSHOT INTEGRATION MATRIX

FIGURE #	SCREENSHOT CAPTION	LIVE APP LOCATION	DISSERTATION EXPLANATION CONTEXT
Figure 4.5	Institutional Multi-Factor Auth Screen	Auth Modal (Sign-In view)	Demonstrates Email/Password and WebAuthn hardware passkey authentication.
Figure 4.6	Privileged Role Passcode Gatekeeping	Auth Modal (Register view)	Demonstrates institutional key requirement (GWU-FACULTY-2026) preventing escalation.
Figure 4.7	Lecturer Real-Time Streaming Radar	Lecturer Portal View	Shows active session timer, WebSocket live streaming radar table, and defaulter rosters.
Figure 4.8	Flagged Exception Resolution Modal	Lecturer "Review & Override"	Illustrates solution to sweaty finger lockout failure with NDPA 2023 audit remark.
Figure 4.9	Class Rep Proctoring Dashboard	Class Rep Portal View	Shows hall headcount radar (42/50 in hall) with strict Separation of Duties lockdown.
Figure 4.10	Classroom Kiosk Scanner Terminal	Kiosk View	Shows optical platen laser sweep, minutiae extraction HUD, and acoustic feedback cues.
Figure 4.11	Student 75% Compliance Gauges	Student Portal View	Illustrates automated statutory NUC percentage computation and deficit forecasts.
Figure 4.12	Official Exam Clearance Docket	Docket Modal View	Shows printable university examination docket with anti-tamper QR security token.
Figure 4.13	Admin Portal & NDPA Audit Logs	Admin Portal View	Demonstrates 1-Click Firestore Cloud Seeder, SQL export, and immutable audit trails.